

Abiram K

SOFTWARE DEVELOPER

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SUMMARY

An independent, self-driven, and hardworking Software Developer with a strong work ethic and a bias toward shipping. Experienced in building AI-first platforms, enterprise-grade cloud architectures, and highly scalable backend solutions. Adept at system design, optimizing legacy architectures, advanced LLM pipeline construction, and production DevOps orchestration.

SKILLS

Languages:	JavaScript, TypeScript, Python, SQL
Frameworks:	React.js, Next.js, React Native (Expo), NestJS, Node.js, Express.js, FastAPI, Redux, Vite, Zustand
Databases & Cache:	PostgreSQL, MySQL, MongoDB, Redis, TypeORM, Prisma ORM
Architecture:	REST APIs, Hexagonal Architecture (Ports & Adapters), Repository Pattern, SOLID Principles, Clean Architecture, Code Optimization
DevOps & Cloud:	Docker, AWS (EC2, S3), Google Cloud, Azure, NGINX, Cloudflare, PM2, GitHub Actions
Integrations:	LLM Autonomous Agents, RAG Pipelines, Whisper AI Speech-to-Text, WebRTC, Socket.IO, Razorpay, Stripe, JWT, OAuth
Tools/Platforms:	Git, Postman, Figma, Touch typing, Data Structures and Algorithms

EXPERIENCE

Annam.AI — Software Developer

June 2025 – Present

- Optimizing legacy codebase and refactoring system logic to dramatically improve overall performance, clean architecture, and long-term maintainability.
- Building and scaling multiple production-grade web and mobile applications from scratch at the intersection of fullstack engineering and robust system design.
- Integrating state-of-the-art AI solutions, autonomous agents, tool usage frameworks, and intelligent LLM/RAG workflows directly into consumer production applications.
- Developing cross-platform mobile applications utilizing React Native (Expo) to deliver high-performance, real-time user experiences.

CORE PRODUCTION PROJECTS

Question Collection Platform — Cloud-Native Agri-Mobile Ecosystem

Annam.AI

A high-throughput, cloud-native system engineered to capture, auto-validate, and process distributed domain data at scale.

- Architected a mobile application layout leveraging **React Native (Expo)** paired with a robust **NestJS** and **FastAPI (Python)** backend structure.
- Implemented intelligent, multilingual text processing pipelines and built an automated AI validation service integrated with distributed caching via **Redis**.
- Designed a data-driven infrastructure utilizing **PostgreSQL (TypeORM)** to manage high-volume submission logs safely and efficiently.
- Containerized ecosystem deployment leveraging **Nginx**, **MinIO** / **AWS S3** object storage, and automated cloud workflows.

Technologies Used: React Native, NestJS, FastAPI, Python, PostgreSQL, TypeORM, Redis, AWS S3, Nginx, Docker

Ajrasakha — AI-Powered Agricultural Advisory Platform

Annam.AI

Built scalable web applications and review workflows that connect farmers with AI-assisted, expert-validated agricultural guidance through multilingual interfaces and knowledge management systems.

- Leading a technical engineering team with a strict focus on robust system design, clean architecture, and reliable production deployments.
- Designed and integrated AI/LLM workflows and interactive expert review loops to handle over 25k+ domain-specific agricultural queries.
- Built responsive UI components using React.js and a highly scalable backend infrastructure.
- Collaborated with cross-functional teams to deliver new features and enhance overall user experience.

ViBe — AI-First E-Learning Platform

Annam.AI

An intelligent e-learning ecosystem focusing on contextually rich, adaptive, and integrity-assured student evaluation tracks.

- **Active Learning via Adaptive Challenges:** Developed custom evaluation loops that continuously assess student comprehension, prompting targeted material reviews whenever thresholds are unmet to guarantee robust mastery before advancement.
- **AI-Enhanced Question Generation:** Integrated advanced algorithmic models to generate contextually relevant, highly adaptive questions that reinforce deep knowledge acquisition.
- **Secure & Integrity-Assured Assessments:** Engineered automated smart proctoring frameworks and real-time engagement verification mechanisms to sustain reliable, non-intrusive monitoring environments.

Technologies Used: React.js, Node.js, Express.js, MongoDB, AI-driven Proctoring, Adaptive Learning Workflows

SPANDAN — Real-Time Presentation Polling Platform

A lightweight, low-latency monorepo system facilitating live response tracking and interactive presentations for classrooms.

- Developed room session states and instant question submission pipelines utilizing **Socket.IO** with a **Zustand** and **React Router** client infrastructure.
- Built an asynchronous speech-to-text layer using local **Whisper AI Transformers** for automated question voice transcription.
- Implemented structured workflows to support role-based capabilities, restricting question creation, response validation, and data dashboard tracking natively between teachers and students.

Technologies Used: React, Vite, Node.js, Express, Socket.IO, MongoDB, Mongoose, Zustand, Transformers (Whisper AI), JWT, TailwindCSS

INDEPENDENT SIDE PROJECTS

BLUE SKY SERVICE HUB – Workforce & Service Management Platform — [GitHub](#)

A scalable full-stack platform designed for operations, staff onboarding, and service delivery coordination.

- Engineered a modular backend utilizing **NestJS** structured via **Hexagonal Architecture (Ports & Adapters)** to maintain a strict separation of domain rules from external adapters.
- Developed a clean, responsive web interface using **Next.js (App Router)** and **Tailwind CSS** to manage service allocations, tracking, and customer onboarding profiles.
- Implemented a schema data storage layer using **SQL (PostgreSQL/MySQL)** with **Prisma ORM** to process job trackers, ESI support calculations, and dynamic role-based operations safely.

Technologies Used: Next.js, NestJS, TypeScript, SQL, Prisma ORM, Tailwind CSS, Docker, JWT, Role-Based Access Control

STACKNEST – AI-Powered Social Platform for Developers — [GitHub](#)

- Architected the backend using **Node.js**, **Express.js**, and **Repository Pattern** adhering strictly to **SOLID principles**.
- Implemented real-time communication infrastructure using **Socket.IO** and **WebRTC** for live voice, video, and collaboration rooms.
- Designed an AI-powered feed system using **Multer** and **Cloudinary** for media delivery, optimized via **Nginx** reverse proxy on **Google Cloud**.

Technologies Used: React.js, TypeScript, Node.js, Express.js, MongoDB, Socket.IO, WebRTC, Cloudinary, Multer, OAuth, Stripe, Razorpay, Google Cloud, Figma

SKILL PULSE – E-Commerce Platform — [GitHub](#)

- Developed a high-throughput modern e-commerce application featuring product listings, active cart management, wishlists, and order pipelines.
- Integrated secure payment gateways including **Razorpay**, **COD**, and **Wallet** systems for flexible transactions.
- Deployed the solution onto **AWS EC2** and **Microsoft Azure** with **PM2** and **NGINX** reverse proxy for high availability.

Technologies Used: Node.js, Express.js, MongoDB, React.js, Tailwind, Razorpay, Stripe, Nodemailer, JWT, Cloudinary, NGINX, AWS EC2

EDUCATION

MERN Stack Development

2024 – Present

Brototype, Ernakulam

BSC Computer Science

2021 – 2024

Chinmaya Mission College, Thrissur **CGPA: 7**